CSA04-Operating System

TASK 4 – Evaluate the Services an Operating System Must Provide

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1. Aim

To evaluate the operating-system services required by different smart hospital environments and identify essential and unnecessary services based on performance, reliability and security.

2. Operating-System Services for Each Environment

OS service	Patient monitors	Nurses' tablets	Doctors' desktops	Database server
User interface	Basic display	Touch interface	GUI	Limited administration interface
Program execution	Essential	Essential	Essential	Essential
I/O operations	Essential	Essential	Essential	Essential
File system	Basic	Essential	Essential	Essential
Communication	Essential	Essential	Essential	Essential
Error detection	Essential	Essential	Essential	Essential
Resource allocation	Essential	Essential	Essential	Essential
Accounting	Basic	Basic	Basic	Essential
Protection and security	Essential	Essential	Essential	Essential

Explanation: Every environment needs basic OS services, but their implementation and priority differ according to the device's purpose.

3. Services That Can Be Removed from Bedside Monitors

Bedside monitors require fast responses and limited memory. Therefore, unnecessary services should be removed.

- Complex graphical user interfaces can be replaced with a basic display.
- Unnecessary background applications can be disabled.
- Printing services are not required.
- Unused networking services can be removed.
- Unnecessary accounting and multimedia services can be disabled.

Justification: Removing unnecessary services reduces memory usage, improves performance and helps the monitor respond quickly to abnormal readings. However, alarm handling, communication, error detection and security must remain.

4. Essential Services for the Database Server

The following services are essential:

1. File management: Stores patient records and medical reports.
2. Communication: Allows multiple users and hospitals to access records.
3. Security: Protects confidential patient information.
4. Error detection: Identifies system and storage failures.
5. Resource allocation: Manages CPU, memory and storage.
6. Accounting: Monitors resource usage and system activities.

Justification: These services ensure patient records remain secure, accurate and continuously accessible.

5. How the OS Protects One Patient's Records from Another User's Programs

The operating system uses protection and security mechanisms to prevent unauthorized access.

1. User authentication: Allows only authorized users to log in.
2. Access permissions: Controls who can read, write or modify patient records.
3. Process isolation: Prevents one program from accessing another program's memory.
4. Encryption: Protects patient information from unauthorized access.
5. File permissions: Restricts access to confidential medical files.

Example: A nurse can view patient records, but only authorized doctors or administrators may modify specific medical information.

6. Services Required in Other Environments

Environment	Important OS services	Justification
MRI imaging server	Program execution, I/O, resource allocation, file management	Supports image processing and large file storage.
Partner hospital network	Communication, security, error detection	Enables safe and reliable record sharing.

7. Alternative Concept and Complexity Analysis

Alternative: Use a full desktop operating system with all services enabled on nurses' tablets instead of a reduced handheld operating system.

Factor	Handheld OS	Full desktop OS
Memory	Requires less RAM	Requires more RAM
Storage	Requires less storage	Requires more storage
Start-up time	Generally faster	Generally slower
Battery life	Better power optimization	May consume more battery
Performance	Optimized for mobile devices	Contains unnecessary background services

Complexity Analysis

Space Complexity:

- Handheld OS requires less memory and storage because unnecessary services are removed.
- Desktop OS requires more memory and storage because it includes additional background services and applications.

Time Complexity:

- Handheld OS generally starts faster because fewer services need to load.
- Desktop OS may take longer to start because more services and processes must be initialized.
- Additional background processes can increase CPU usage and reduce battery life.

Evaluation: A full desktop OS is not suitable for nurses' tablets because it consumes more resources and may reduce battery life. A reduced handheld OS is the better choice.

8. Final Evaluation

Environment	Recommended OS services
Patient monitors	Basic interface, program execution, I/O, communication, error detection and security
Nurses' tablets	Touch interface, file management, communication and power management
Doctors' desktops	GUI, multitasking, file management and security
Database server	File management, communication, protection, accounting and error detection
MRI imaging server	Resource allocation, I/O, file management and program execution
Partner hospital network	Communication, authentication, encryption and error detection

9. Result

Thus, the operating-system services required by different hospital environments were evaluated. Unnecessary services should be removed from bedside monitors and nurses'

tablets to improve performance and efficiency, while security, communication, error detection and resource management must be maintained.